



الجامعة الأوروبية بـفاس
EURO-MEDITERRANEAN UNIVERSITY OF FES
UNIVERSITÉ EURO-MÉDITERRANÉENNE DE FÈS

LIBRARY SYSTEM MANAGEMENT

Realized by:

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INTRODUCTION:

Library system management streamlines library operations using technology, making them efficient, user-friendly, and adaptable for modern needs like digital resources and self-service portals.



Question Asked:

*How does the Library Management System improve the operations of the UEMF library compared to traditional methods?



Main Idea

The Library Management System replaces traditional manual processes like handwritten logs and physical cataloging with a streamlined, automated approach. This transition enhances efficiency, reduces errors, and improves user satisfaction, making it an essential tool for modern library operations.

Plan:

1. presentation of the project
2. specification of needs
3. design of the solution
4. realization of the application

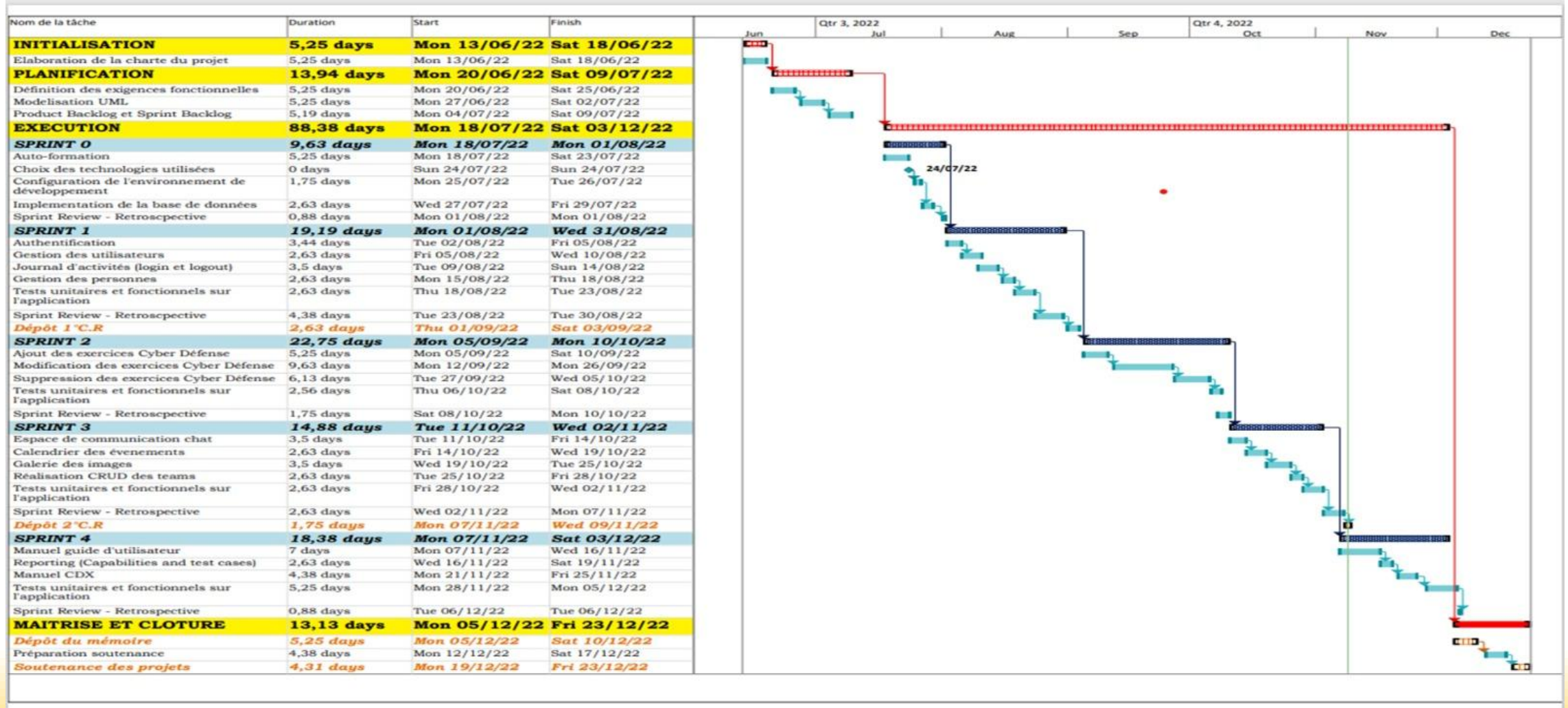


1- Presentation Of The Project :

1-1 General context and objectives



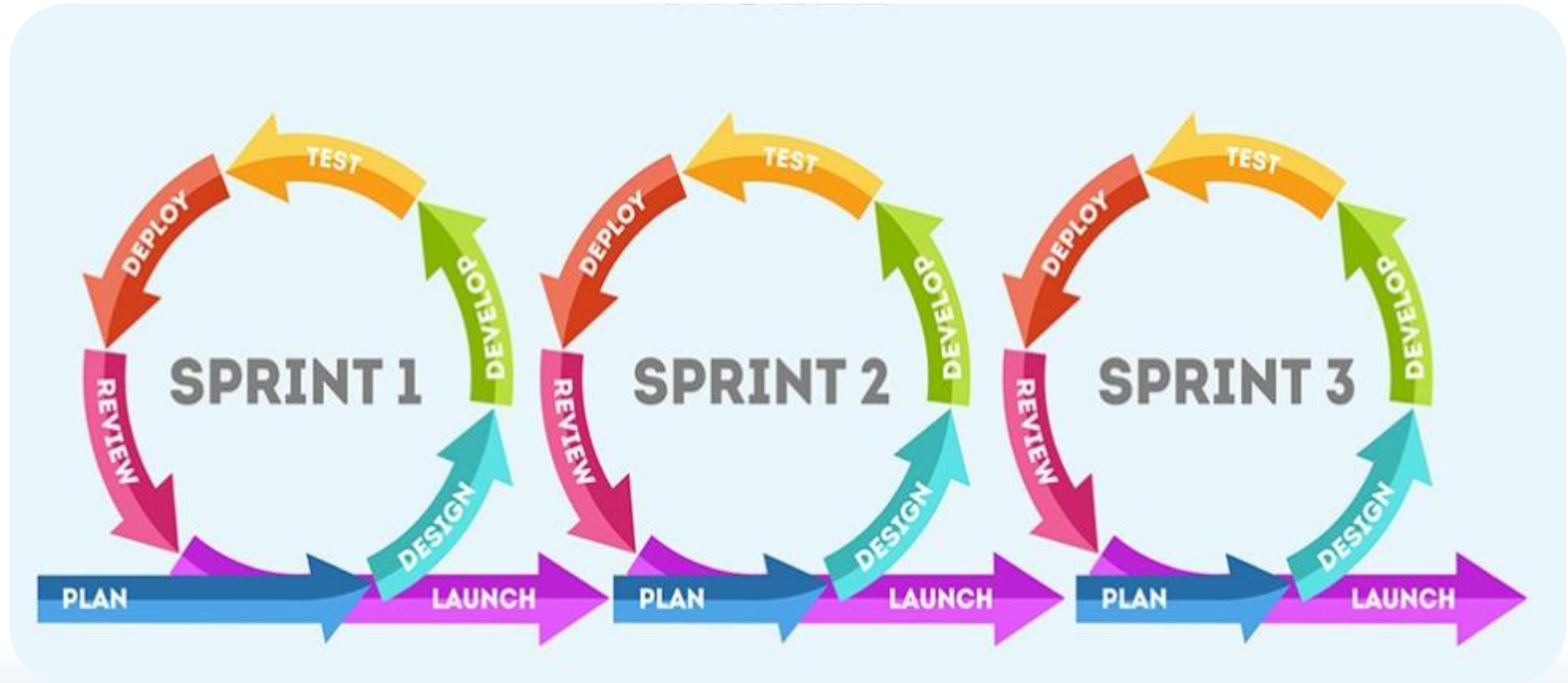
1-2- Project Management And Planning



1-3-Development Methodology



1-3-Development Methodology



2- Specification Of Needs

2-1 Functional requirements :

Server-Level Functionalities:

- The central server manages operations and stores data securely in a local database.
- Data updates are synchronized locally, ensuring accuracy and consistency on the user's machine.



2- 2-Functional requirements :

Client-Level Functionalities:

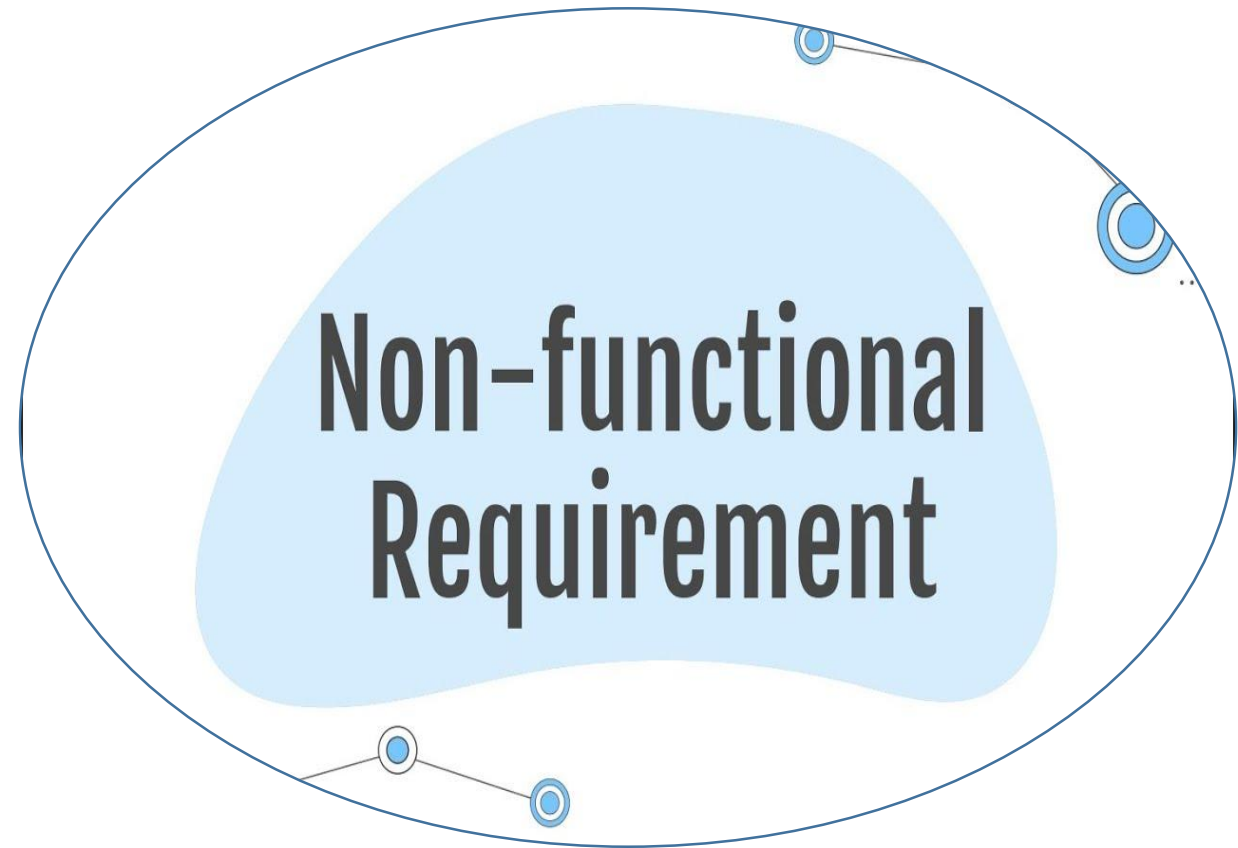
- Users access the desktop application locally, requiring network connectivity only for database or server access.
- Each user is assigned a unique identifier, ensuring secure access to the application and allowing personalized user functionality.
- Authentication with valid credentials is required before users can perform any operations, ensuring that only authorized individuals can access and modify data.

Client-Level Functionalities:

- The system ensures that all changes are reflected locally, meaning data updates are maintained and displayed in real-time on the user's device.
- A user account cannot be accessed from multiple devices simultaneously to prevent data conflicts or inconsistencies.
- The system provides an intuitive interface for Admins and Librarians to efficiently manage library records, including adding, viewing, and issuing books.

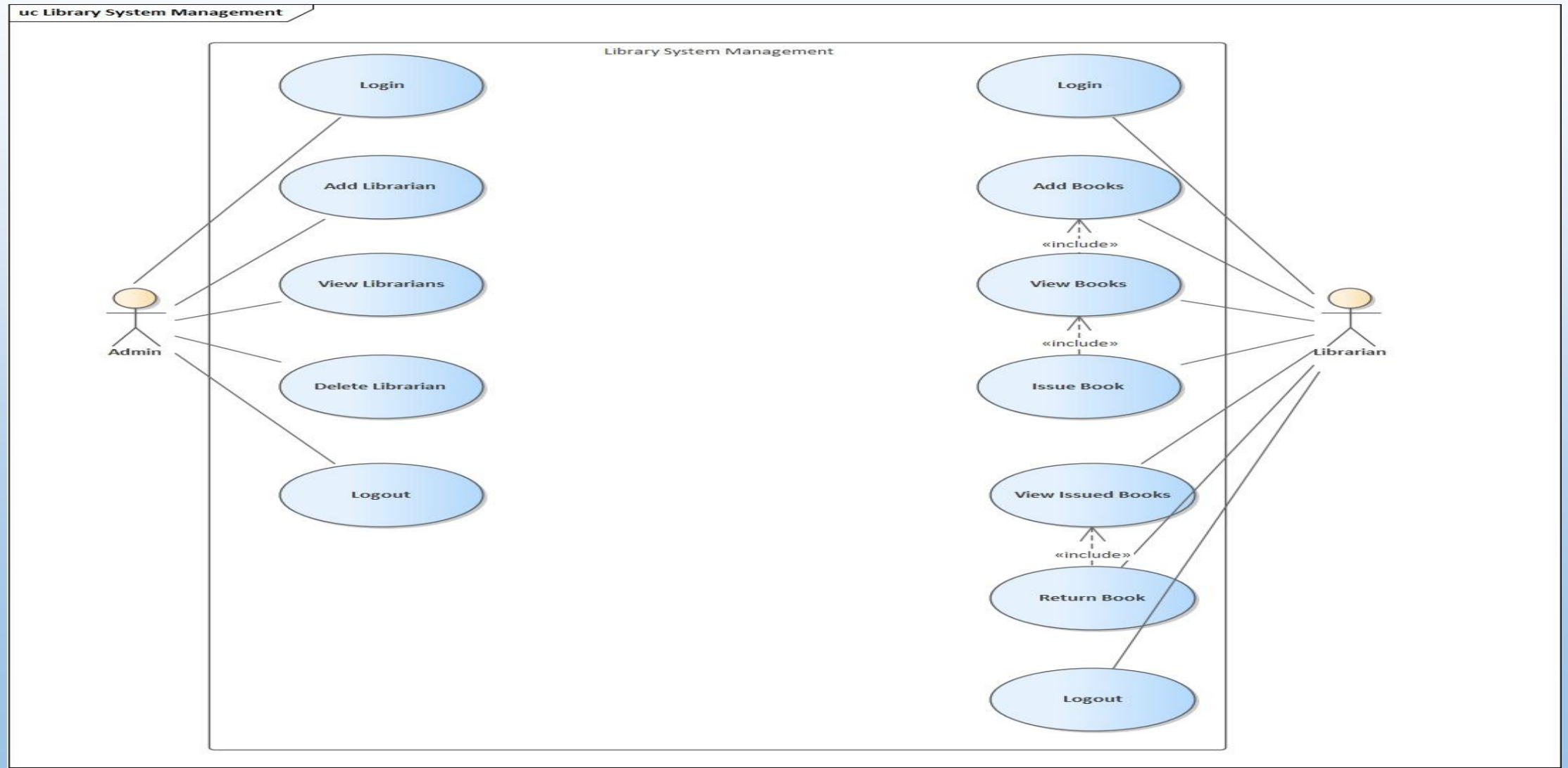
2- 2-Non Functional Requirements :

- **Performance**
- **Scalability**
- **Reliability**
- **Security**
- **Usability**
- **Maintainability**
- **Availability**

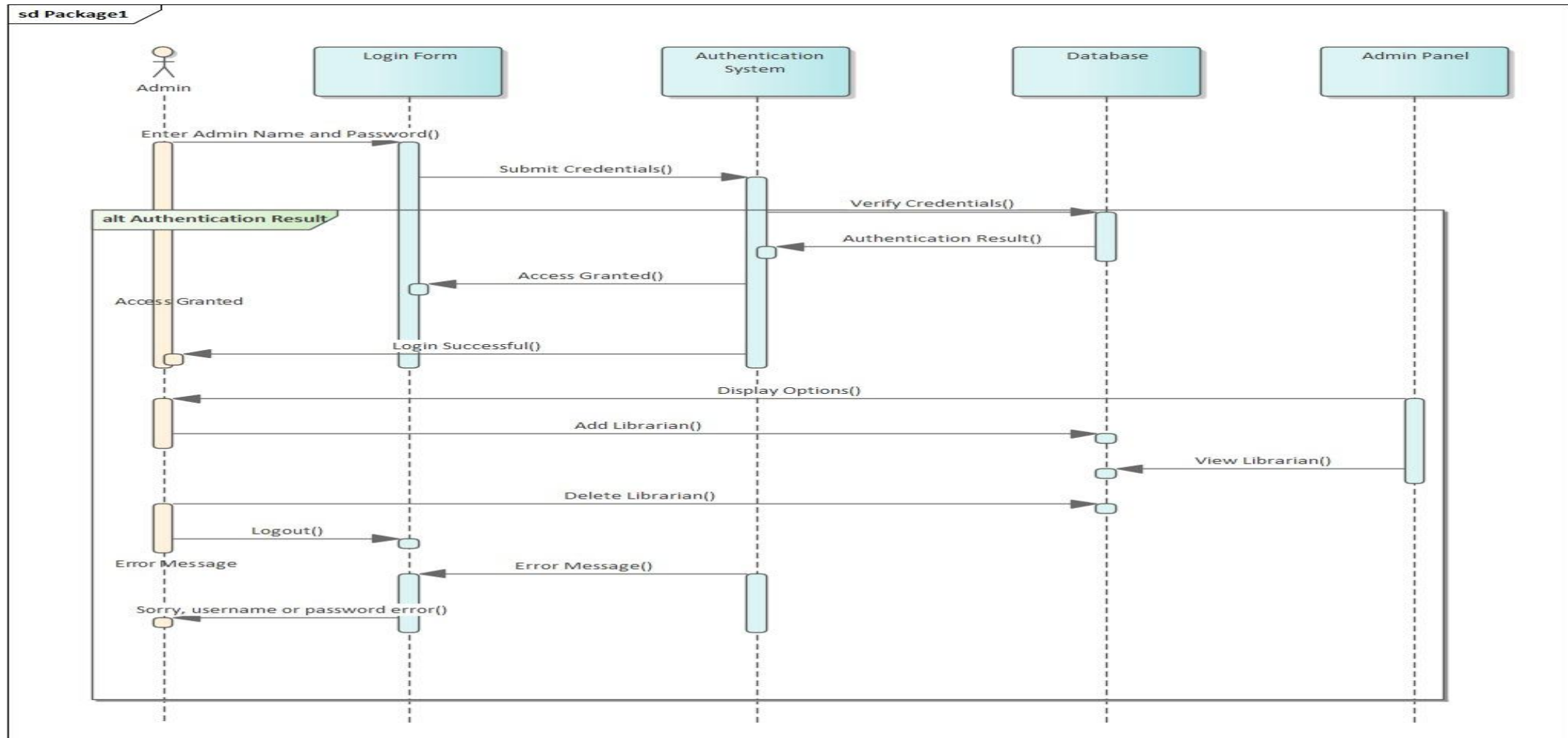


3- Design Of The Solution

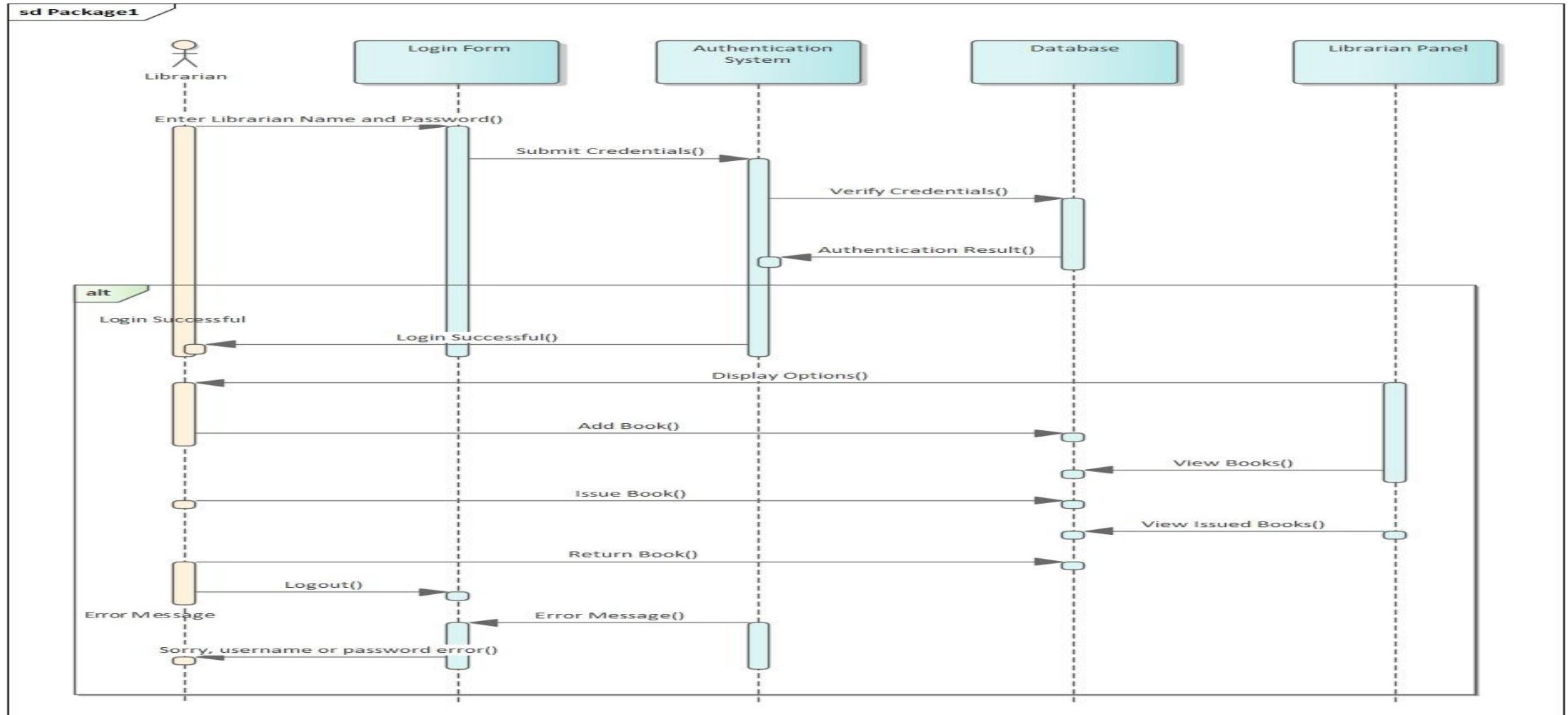
3-1 Use case diagramme



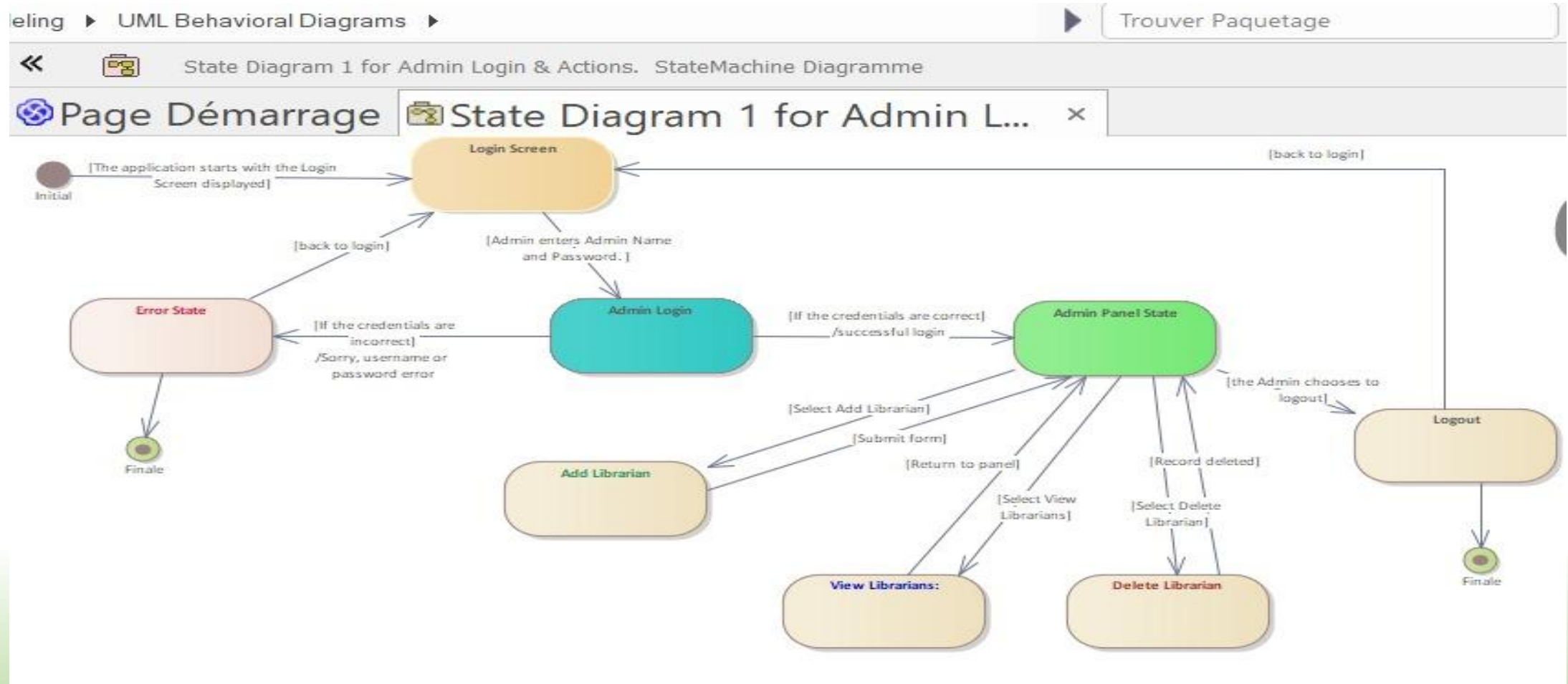
3-1- Sequence Diagram (Admin)



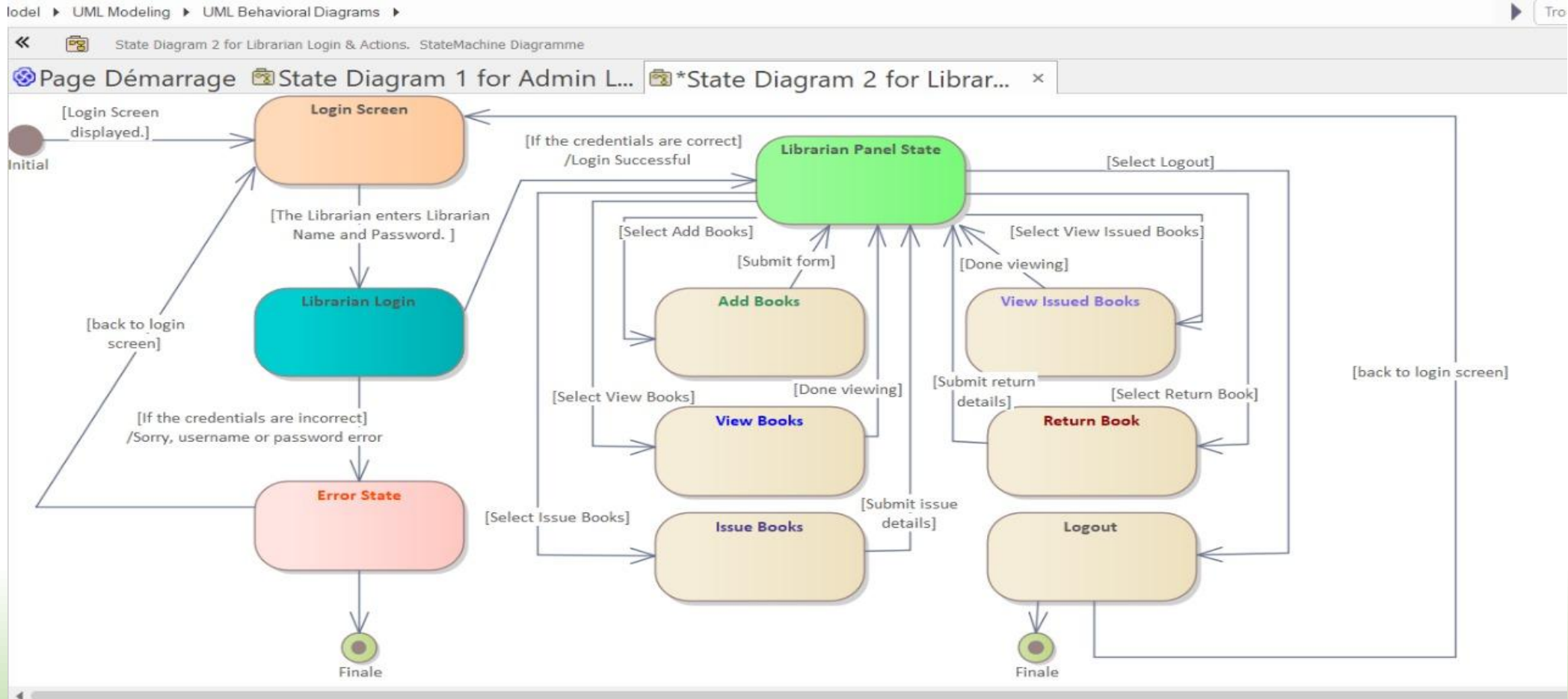
3-2- Sequence Diagram (Librarian)



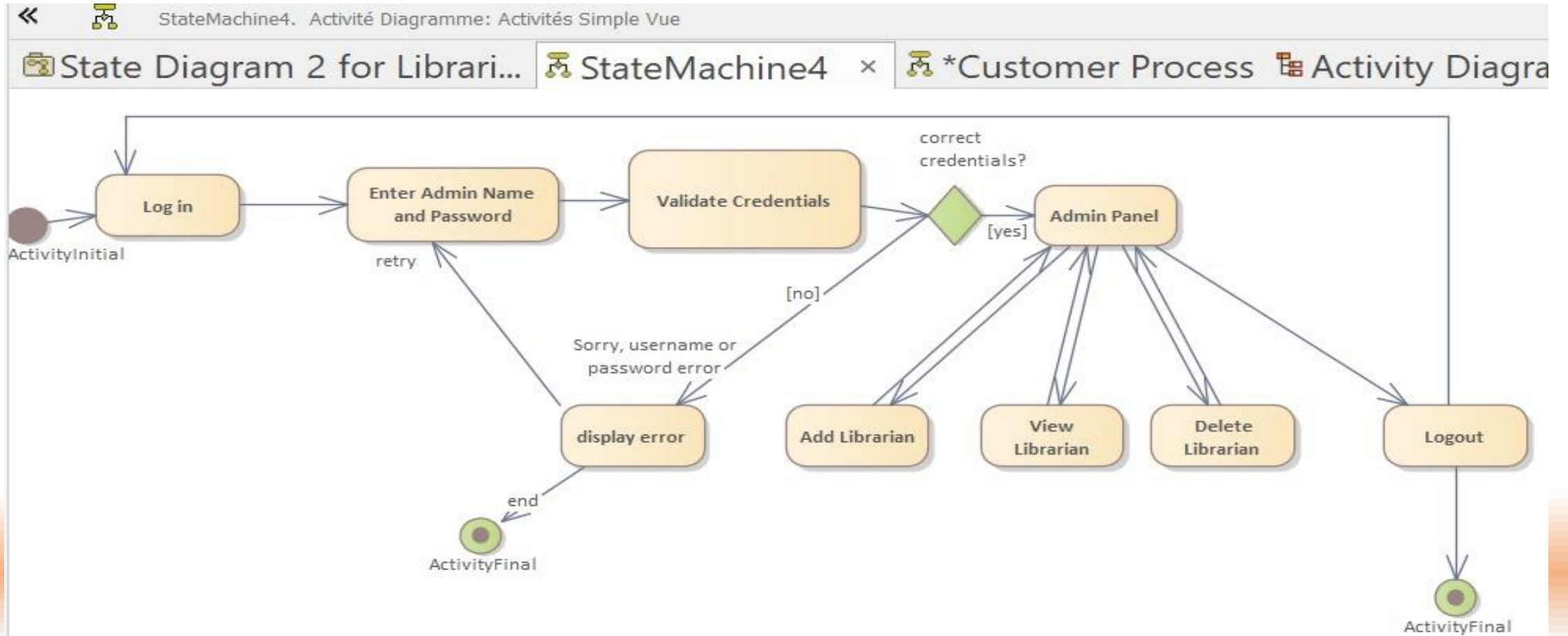
3-3 State Diagram (Admin)



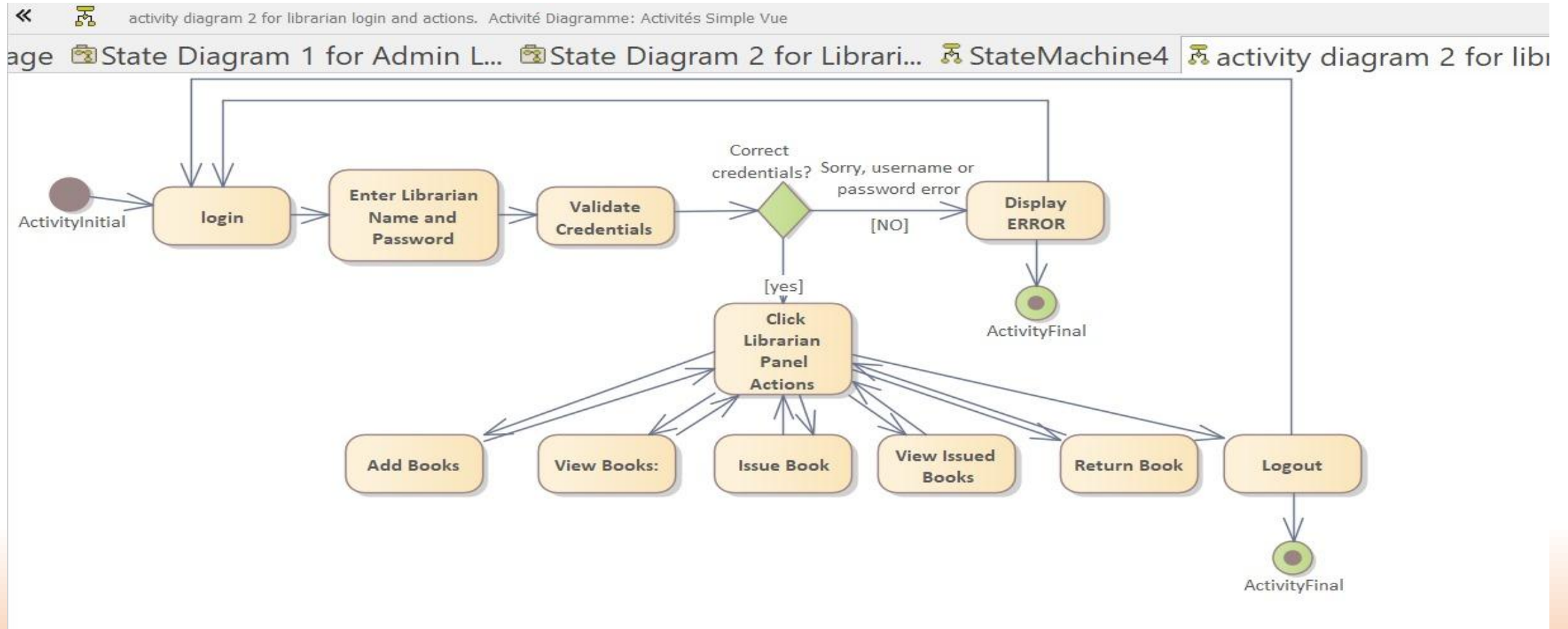
3-3 State Diagram (Librarian)



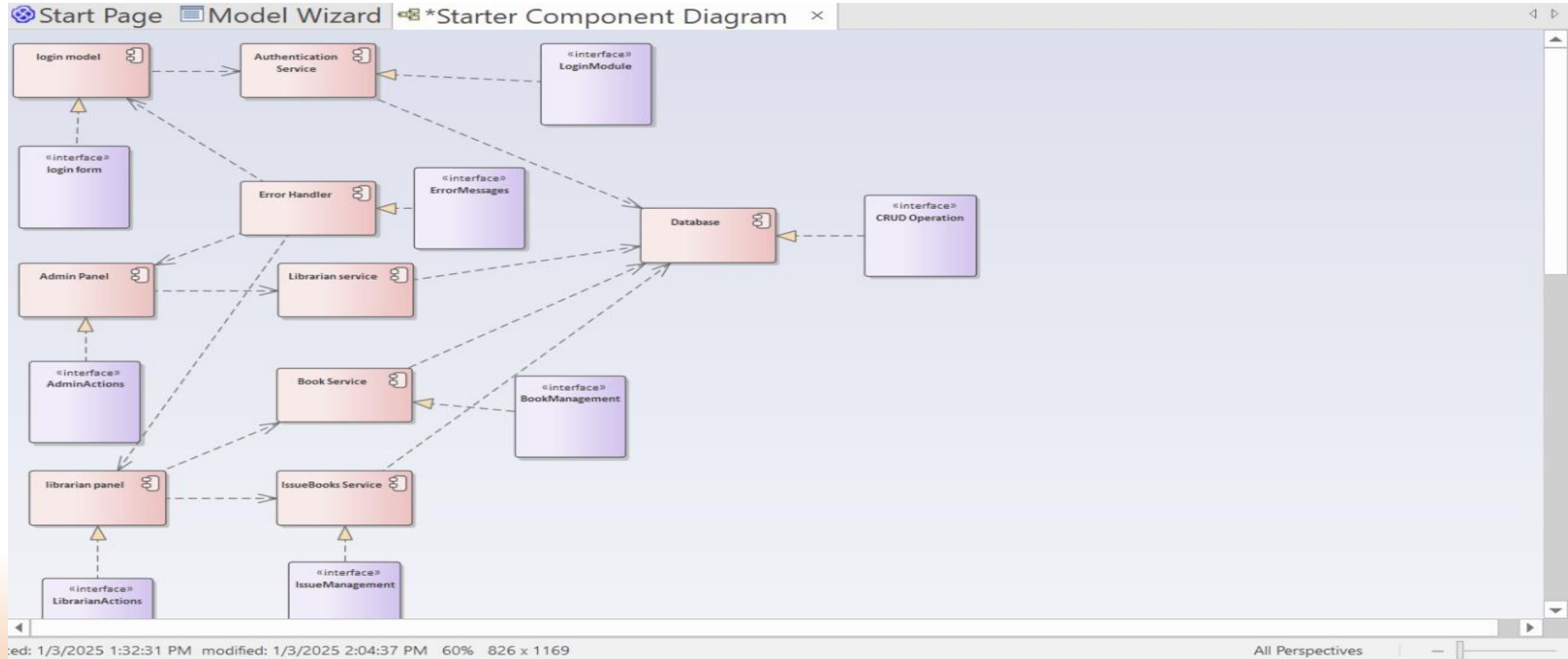
3-4 Activity Diagram (Admin)



3-4 Activity Diagram (Librarian)



3-5 Component Diagram



4- Realization Of The Application

- **Software Engineering Lab:**



- **SGBD :**



- **IDE :**



4-2- Demonstration



Conclusion:

- In conclusion, the Library Management System desktop application offers an efficient and user-friendly solution for managing library operations. It simplifies the tasks for both Admin and Librarian roles, making it easy to manage books, transactions, and user data. The system ensures smooth and reliable operations with real-time data synchronization, supporting seamless interactions across all functions.
- With its intuitive interface and effective role-based features, the system provides a solid foundation for library management, ensuring streamlined processes and adaptability for future needs.